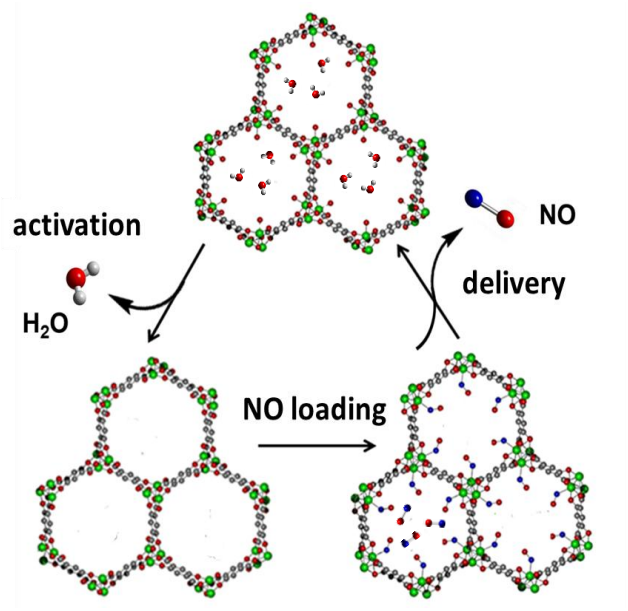
Simulating Nitric Oxide Diffusion within MOF-based Polymers used to Prevent Catheter Related Thrombosis and Infection



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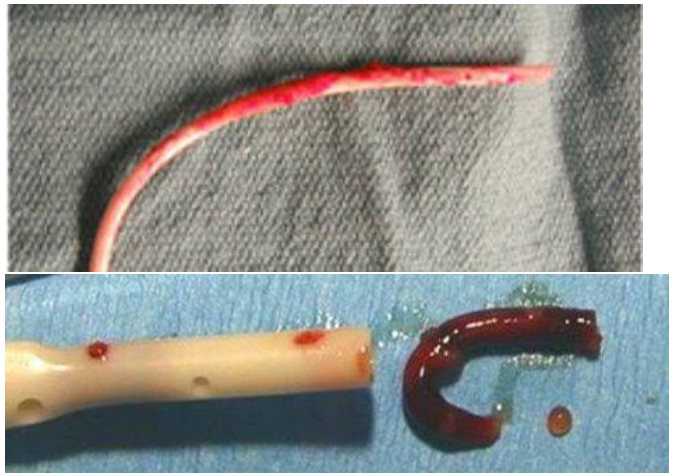
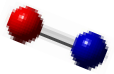
Introduction : Antithrombosis Effect of NO

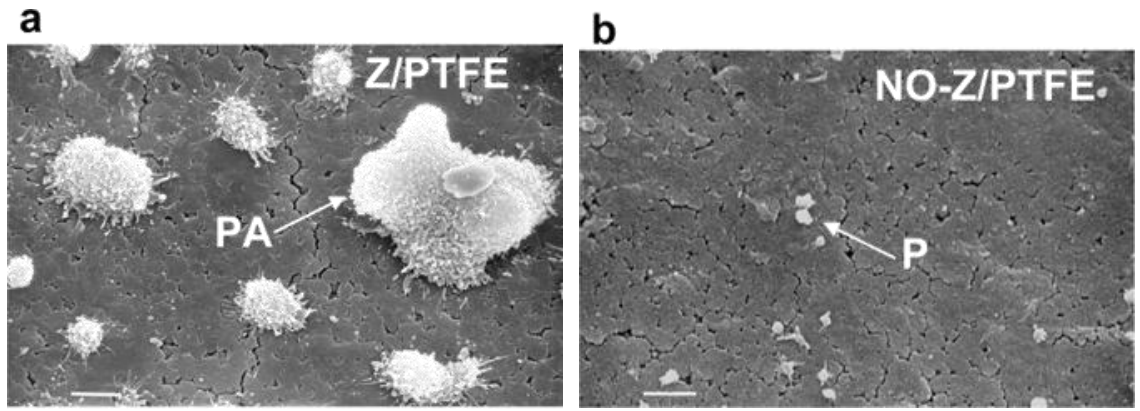
Thrombus & Infection during to Catheterization



- Formation of biofilm on Catheter surfaces
- Long term contact with Catheter
- Coarse Catheter surface
- Shear force while catheter insertion in Bloodstream

Introduction of Nitric Oxide

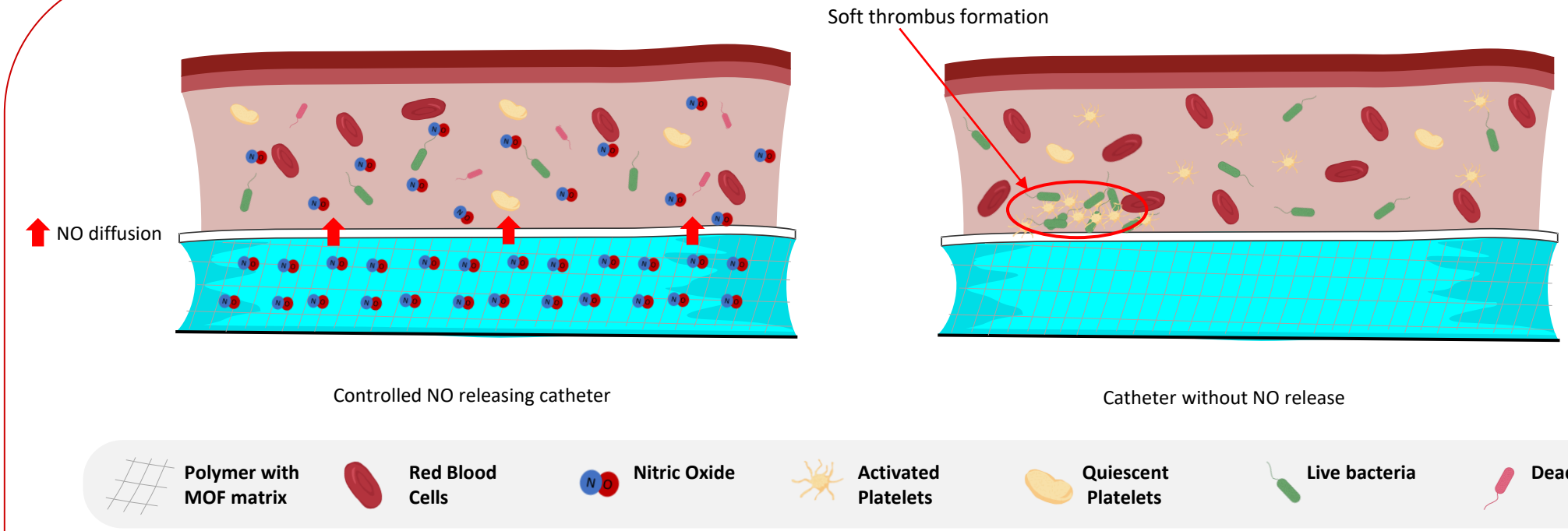




Scanning electron micrographs of the surface of (a) untreated-zeolite-A/PTFE disks (Z/PTFE), (b) NO-loaded Co-zeolite-A/PTFE. [1]

- Stops Coagulation of Inflammatory cells
- Potent inhibitor of Platelet Activation and Adhesion
- Natural Antimicrobial agent in the Immune system
- Kills live Bacteria

Introduction : Antithrombosis Effect of NO

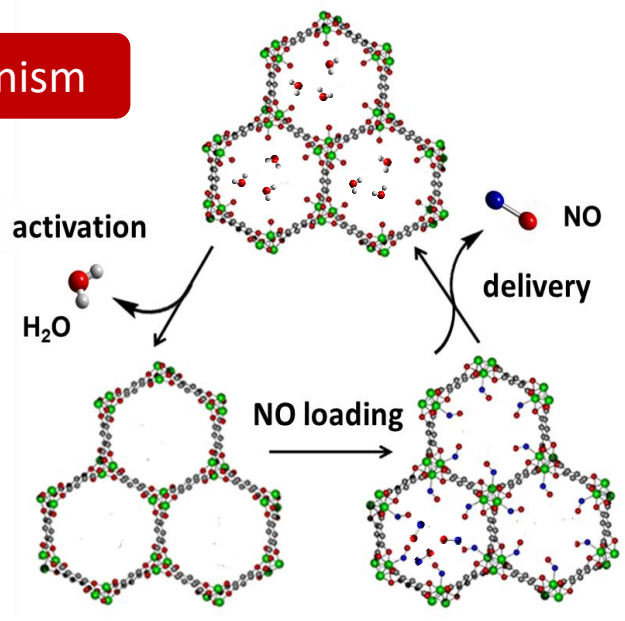


- Activated platelets aggregation causes thrombus formation
- Nitric Oxide in low levels inhibits growth of bacteria and platelets activation
- Nitric oxide prevents activation of platelets, that ultimately leads to thrombus formation
- Excess Nitric Oxide Concentration is Toxic

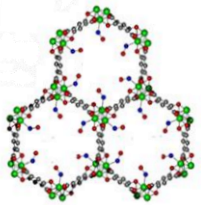
Introduction : MOF(Metal Organic Frameworks) manufacturing and NO loading

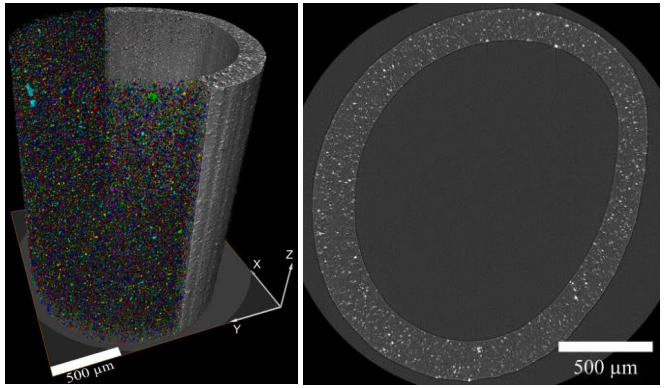
NO Delivery Mechanism

- Loading of NO into MOF
- Delivery of NO into target
- Activation of MOF via H2O removal



MOF micro-CT





Manufacturing

Tube Dimensions:1.75 mm OD with 1.25mm ID

1,2,5 % wt. MOF-polymer blending

Introduction : Project Summary

QUEEN'S UNIVERSITY BELFAST

poly M2F

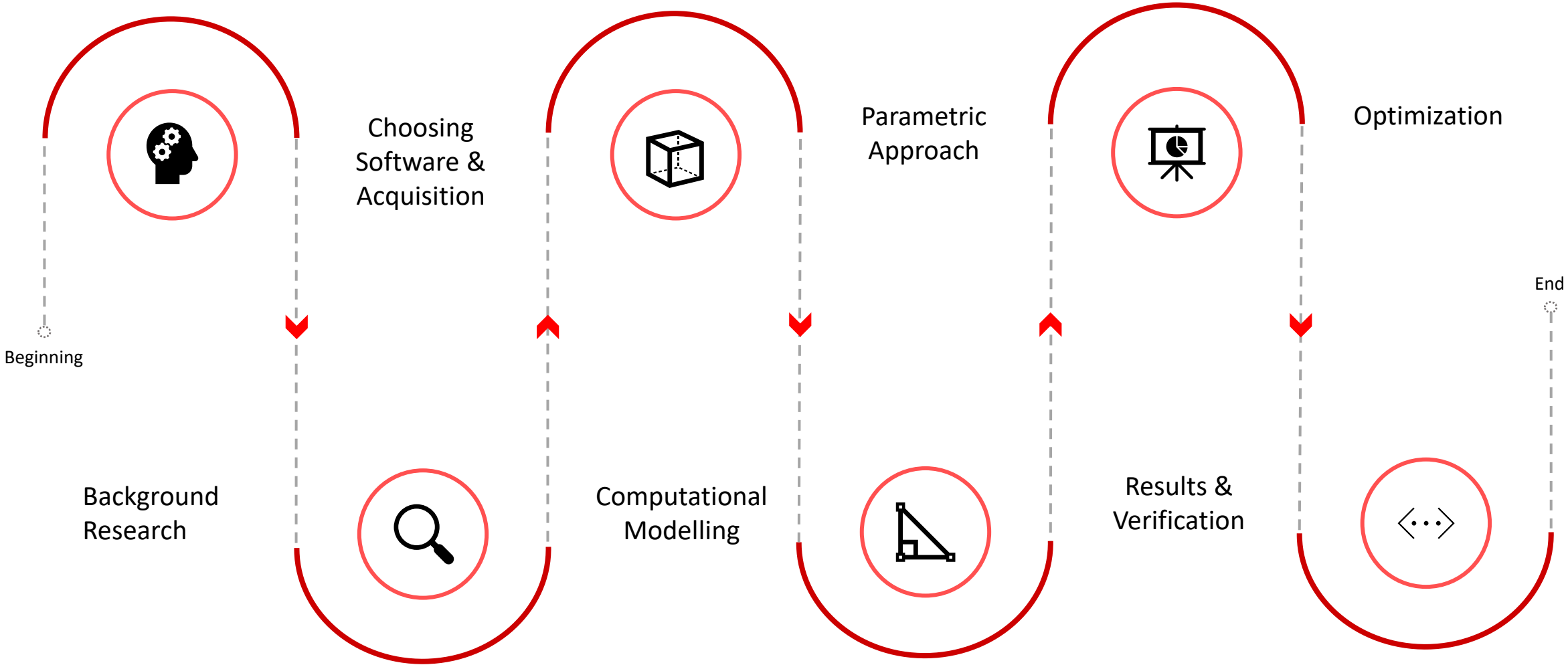
AIM

The project will develop a novel computational model to mitigate problems for catheter related Thrombosis and Infection by Surrogate model, experimental verification and validation for MOF based polymer as material substitute in a Catheter.

OBJECTIVES

1. Multiphase spatio-temporal modelling to predict and tune NO release.
2. Prediction of optimised MOF loading polymer matrix combination to give a desired NO release.
3. NO and water vapour transmission experiments and computational modelling of NO release with verification.
4. Pre-prototype development for implementation in Medical Device.

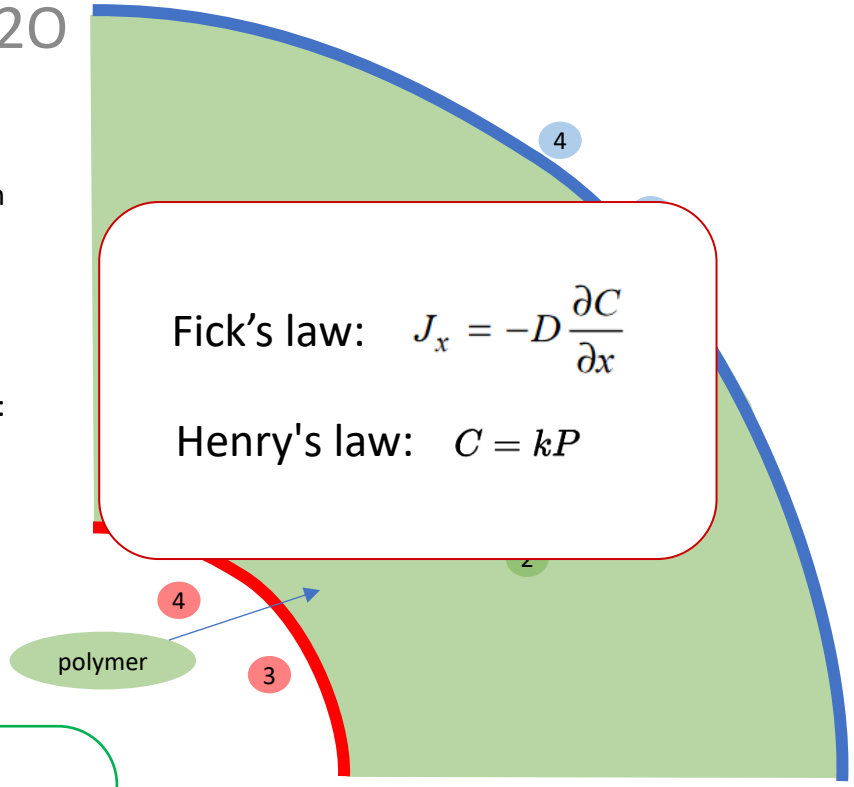
Introduction : Modelling Roadmap



Diffusion Results: Diffusion Model in FEBio

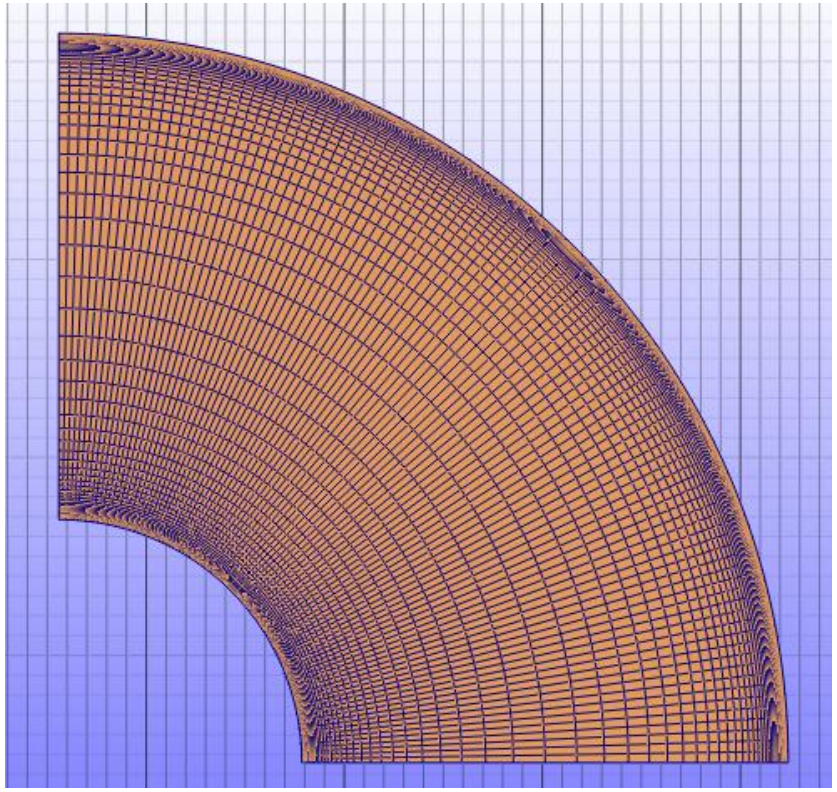
With Reaction, NO & H2O Initial Conditions

1. Initial NO_I(Solid Bounded) concentration in polymer. Conc: 5 mM(nmol/mm²s).
2. Zero Displacement BC on all nodes.
3. Prescribed concentration BC of H2O. Conc: 5 mM.
4. Zero Concentration BC of NO_R.



Fick's law: $J_x = -D\frac{\partial C}{\partial x}$

Henry's law: $C = kP$



(Material + NO_I)+ H2O

⬇

(Material + H2O) + NO_R

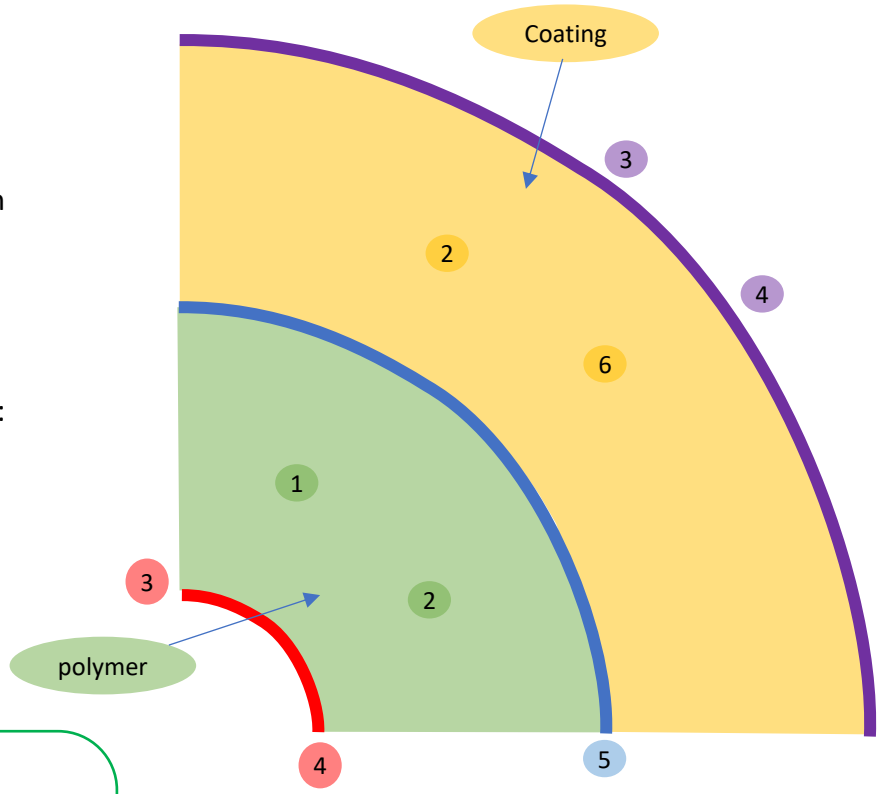
Explanation: NO gas that is loaded within the material interacts with the ingression of H2O and is consumed to form NO releasing (Which can diffuse outwards).

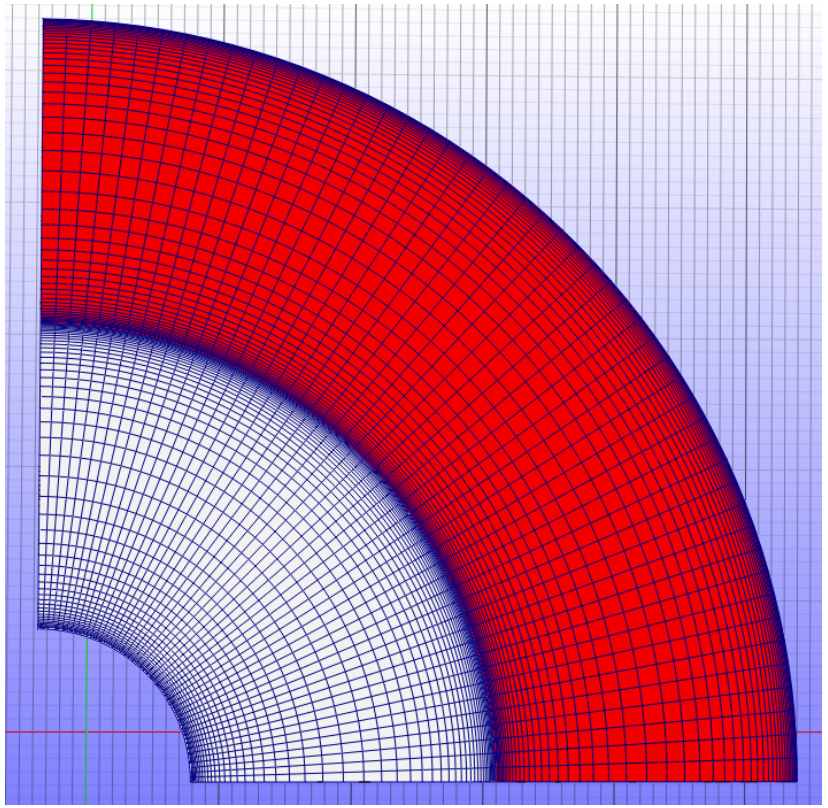
1. NO_I :- The solute acting as NO gas loaded into the material
2. H_2O:- Species works as solute that diffuses into the polymer to activate NO release
3. NO_R :- After activation NO diffuses outwards from the material towards the sink

Diffusion Results: Diffusion Model in FEBio

With Reaction, Contact with Coating

1. Initial NO_I(Solid Bounded) concentration in polymer. Conc: 5 mM(nmol/mm²s).
2. Zero Displacement BC on all nodes.
3. Prescribed concentration BC of H2O. Conc: 5 mM.
4. Zero Concentration BC of NO_R(Releasing NO).
5. Contact Solute Transfer Boundary Layer





(Material + NO_I)+ H2O

⬇

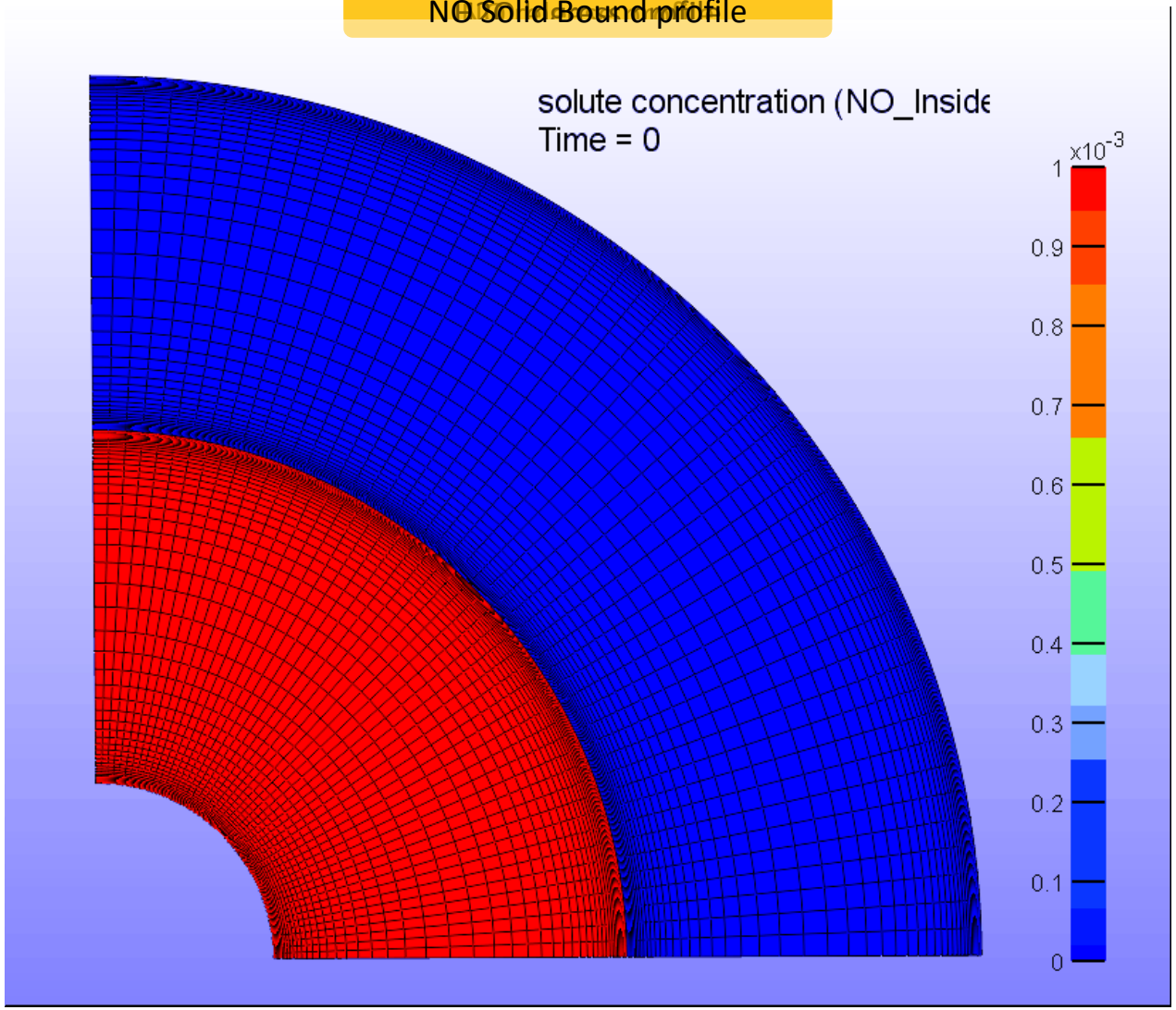
(Material + H2O) + NO_R

Explanation: NO gas that is loaded within the material interacts with the ingression of H2O and is consumed to form NO releasing (this one can diffuse outwards).

1. NO_I :- The solute acting as NO gas loaded into the material
2. H_2O:- Species works as solute that diffuses into the polymer to activate NO release
3. NO_R :- After activation NO diffuses outwards from the material towards the sink

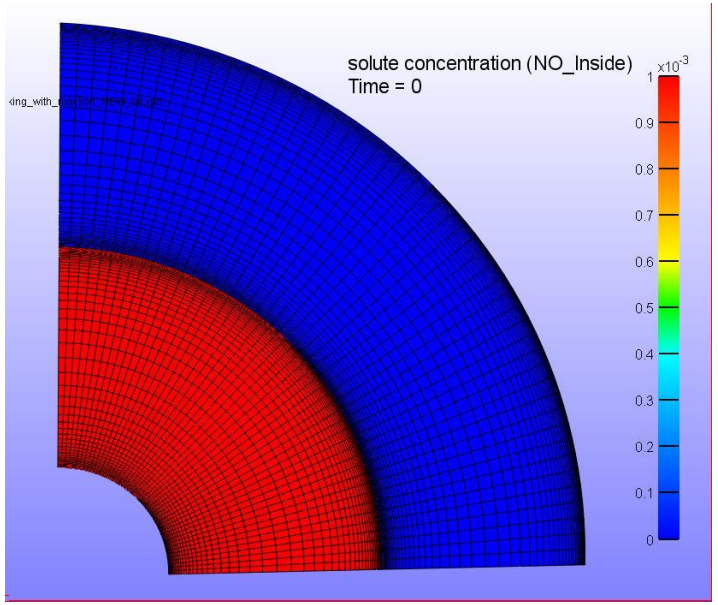
Diffusion Results: Diffusion model in FEBio

NO Solid Bound profile

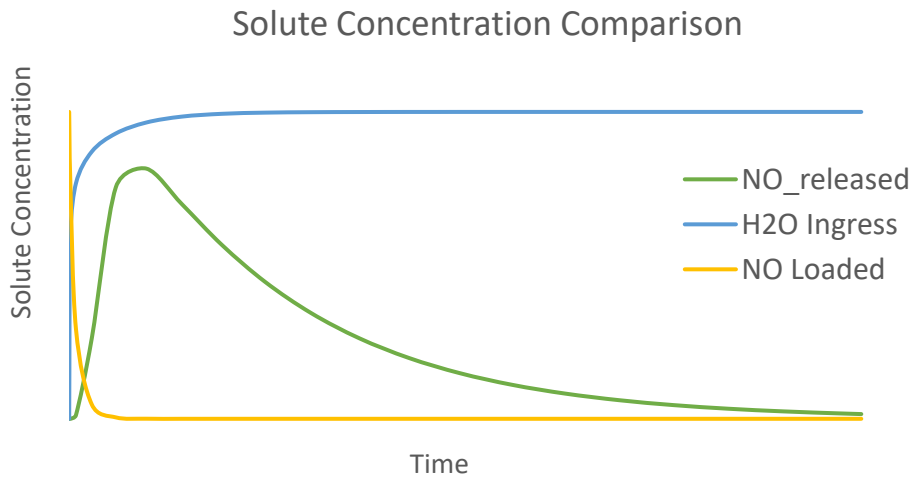


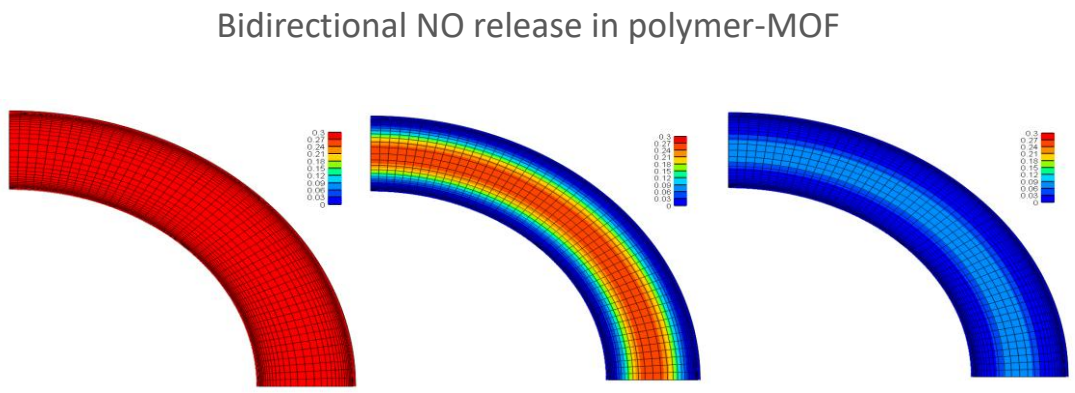
Mesh	A Bi-directional mesh with a bias towards the area of interest
Diffusion	Discussion analysis within a quad-Cylindrical polymer solid geometry with NO Diffusion Coeff = 1e-5 $mm^2\ s^{-1}$(Assumed value)
Solute	A solute(NO) of 0 charge and neutral reactivity at room temperature is analyzed.
Study parameter	Concentration, pressure boundary condition and zero displacement BC at surfaces
Result	For 50000s-time steps, solute concentration gradient, flux gradient.

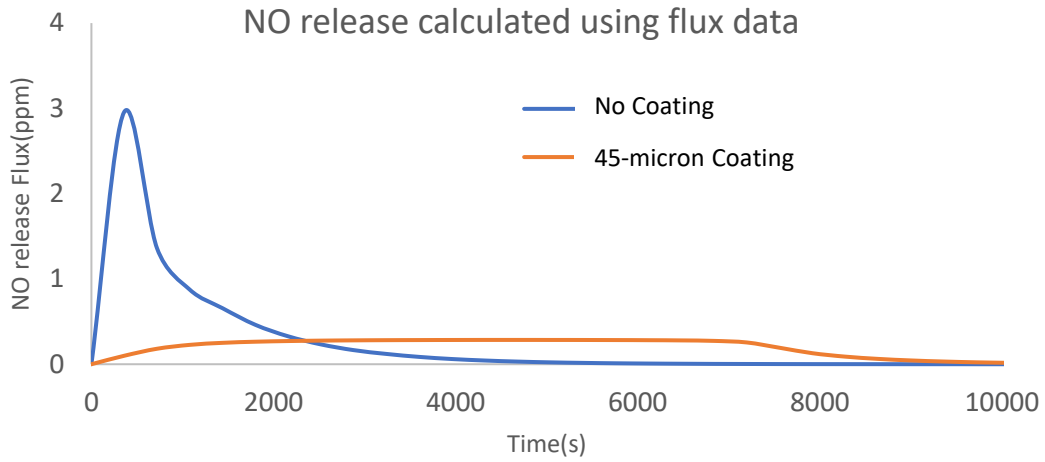
@ timestep 0

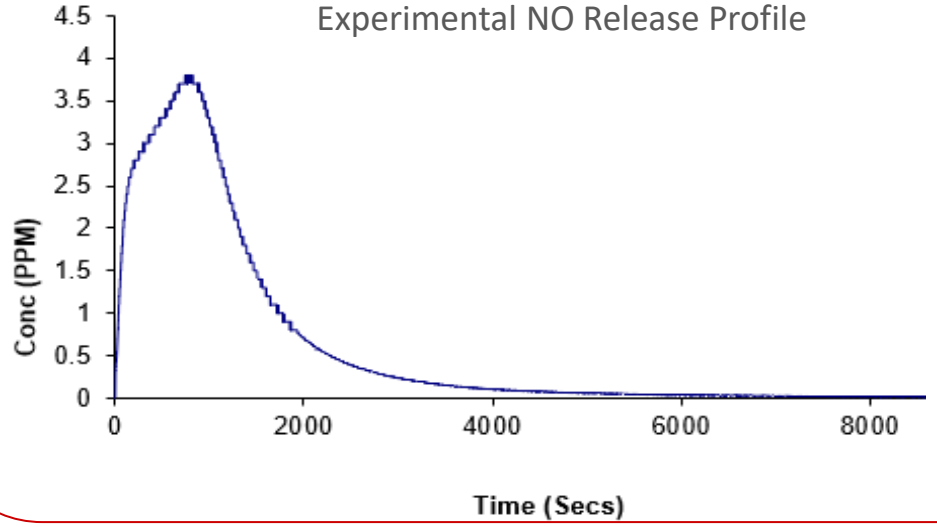


Diffusion Results: Comparison of NO release plots

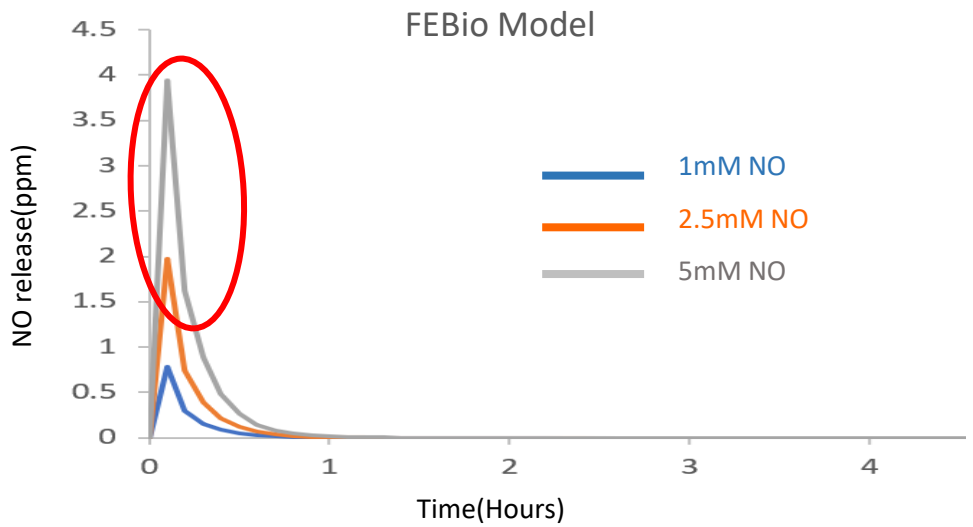


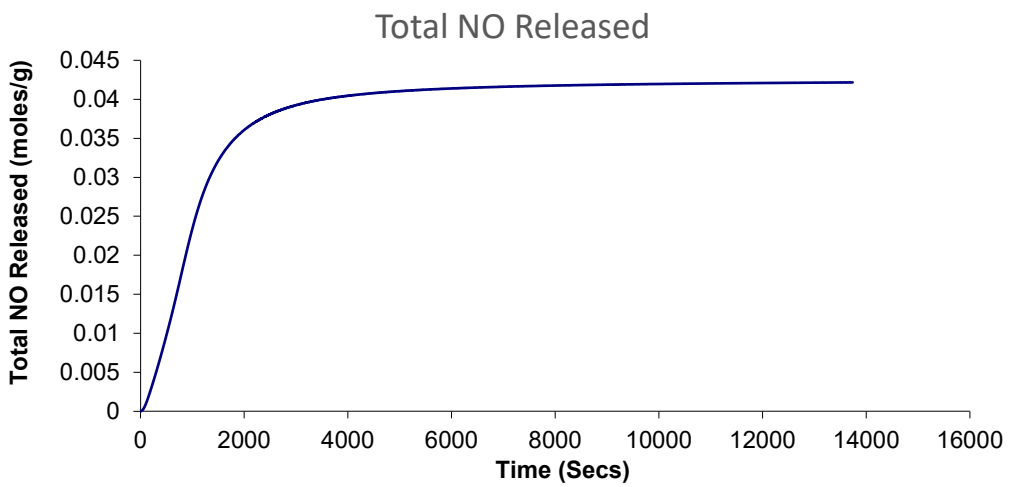






Diffusion Results: Parametric study





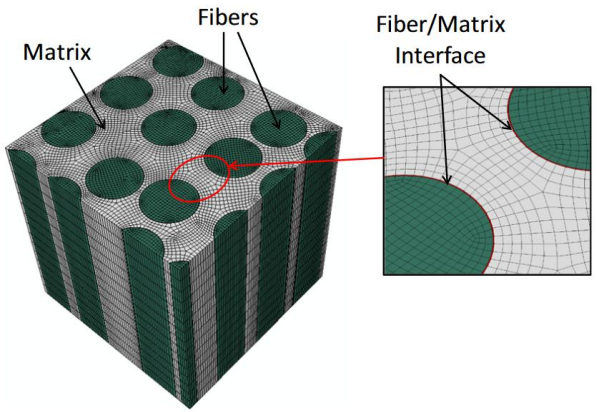
Steady NO Release

- ∝ Coating Thickness
- ∝ Initial MOF Loading
- ∝ Solute Relative Diffusion Coefficient
- ∝ Water Permeability & Solubility in Polymer
- ∝ Polymer Layering?

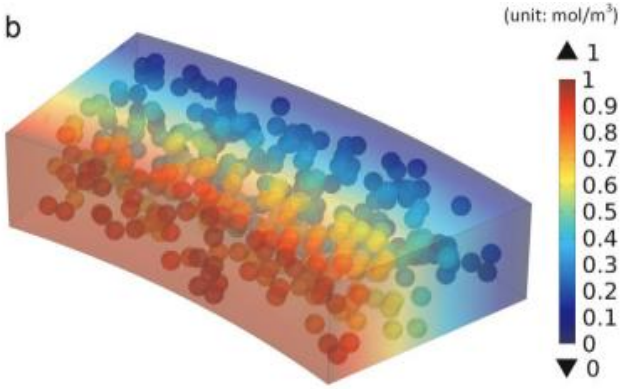
Discussion & Prospectives: Future Prospective

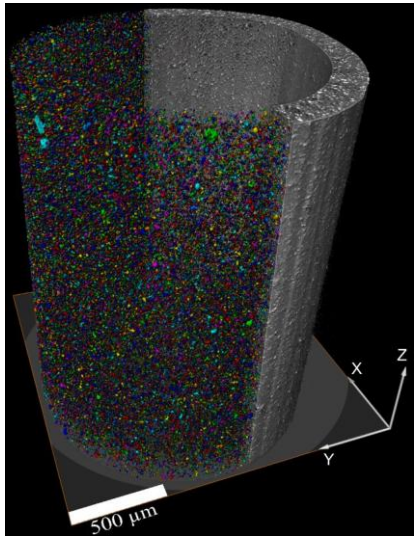
Continuum Model Development → Redefined Model Based on Experimental Data → Simulating Large Dataset → Updated Surrogate Model → Final result Verification and Model Validation

NO Diffusion Tests | NO Release Tests | H2O Diffusion Tests



RVE(Representative volume element) model





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Floor is open for Questions and Discussions.

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